

ENTERPRISE CASE STUDY SUBMISSION

TradeFlow ERP + CRM Operations Portal

Comprehensive Technical Blueprint, System Architecture, API Contract & Operational Specifications

1. Executive Summary & Business Context

TradeFlow Operations Portal is a production-ready full-stack Mini ERP and CRM system built for wholesale and distribution enterprises. The platform integrates sales management, customer CRM tracking, real-time inventory guardrails, purchase order fulfillment, and financial invoicing into an executive dark-mode dashboard.

- **Sales & CRM:** Customer lead lifecycle management, classification (Retail, Wholesale, Distributor), and timestamped follow-up interaction history.
- **Warehouse & Inventory:** Real-time stock tracking, minimum stock alert triggers, and immutable stock movement logging (IN/OUT).
- **Sales Challans Engine:** Automatic numbering (CH-YYYYMMDD-XXXX), immutable product snapshotting, and atomic stock validation preventing negative stock.
- **Accounts & Finance:** Supplier purchase order tracking and customer billing invoice creation.

2. Live System Deployment Matrix

Component	Hosting Platform	Live URL / Reference
Source Code	GitHub Cloud	https://github.com/kotagirisrilaxmi06/trade-flow-
Frontend Web App	Vercel Static Hosting	https://trade-flow-cnc8oix2v-kotagirisrilaxmi06s-projects.vercel.app
Backend REST API	Vercel Serverless Functions	https://trade-flow-cnc8oix2v-kotagirisrilaxmi06s-projects.vercel.app/health
Database Engine	Neon Cloud PostgreSQL	PostgreSQL 18.4 (AWS ap-southeast-1)

Role-Based Access Credentials

Role	Email Address	Password	Permissions Scope
Admin	admin@example.com	password123	Full System Access Across All Modules
Sales	sales@example.com	password123	Customer CRM, Follow-ups & Challans Generation
Warehouse	warehouse@example.com	password123	Product Stock Catalog & Stock IN/OUT Movements
Accounts	accounts@example.com	password123	Financial Invoices & Purchase Order Tracking

3. System Architecture & Tech Stack

The system utilizes a **Unified Monorepo Architecture** deployed on Vercel's serverless infrastructure with an SSL-encrypted PostgreSQL backend.

- **Frontend:** React 18, TypeScript, Vite, CSS with glassmorphism aesthetics, brand logo, and smooth micro-animations.
- **Serverless API:** Node.js, TypeScript, Express.js deployed to Vercel Serverless Functions at `/api/index.ts`.
- **Database Layer:** Neon Cloud PostgreSQL (PostgreSQL 18.4) with Node `pg.Pool` connection pooling.

4. Database Schema DDL

```
-- Users Table
CREATE TABLE IF NOT EXISTS users (
  id TEXT PRIMARY KEY,
  name TEXT NOT NULL,
  email TEXT UNIQUE NOT NULL,
  password TEXT NOT NULL,
  role TEXT NOT NULL
);

-- Customers Table
CREATE TABLE IF NOT EXISTS customers (
  id TEXT PRIMARY KEY,
  data JSONB NOT NULL
);

-- Products Table
CREATE TABLE IF NOT EXISTS products (
  id TEXT PRIMARY KEY,
  data JSONB NOT NULL
);

-- Challans Table
CREATE TABLE IF NOT EXISTS challans (
  id TEXT PRIMARY KEY,
  data JSONB NOT NULL
);

-- Purchase Orders Table
CREATE TABLE IF NOT EXISTS purchase_orders (
```

```

    id TEXT PRIMARY KEY,
    data JSONB NOT NULL
);

-- Invoices Table
CREATE TABLE IF NOT EXISTS invoices (
    id TEXT PRIMARY KEY,
    data JSONB NOT NULL
);

```

5. REST API Technical Specifications

Method	Endpoint	Description	Status Codes
GET	/health	System Health Status Check	200 OK
POST	/auth/login	User Login Authentication	200, 400, 401
GET	/customers	List All Customer Profiles	200 OK
POST	/customers	Register New Customer	201 Created, 400
POST	/customers/:id/follow-ups	Add Customer Follow-up Note	201 Created, 400
GET	/products	List Product Catalog & Stock	200 OK
POST	/products	Add Product to Catalog	201 Created, 400
POST	/products/:id/stock-movements	Log Manual Stock Adjustment (IN/OUT)	201 Created, 400
GET	/challans	List Issued Sales Challans	200 OK
POST	/challans	Issue Sales Challan with Atomic Stock Check	201 Created, 400
GET	/invoices	List Billing Invoices	200 OK
POST	/invoices	Issue Billing Invoice	201 Created, 400